Pamella Soares de Sousa, PhD

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RESEARCH INTERESTS

Blockchain-oriented software engineering; developer experience; empirical software engineering and mining software repositories; NLP and LLM-assisted analysis of software engineering data; quality and security of smart contracts.

EDUCATION

PhD in Computer Science (Software Engineering), State University of Ceará (UECE), Brazil — Mar 2022 - Sep 2026

- Thesis: *A Multidimensional Framework for Blockchain Developer Experience*. Defended 1 September 2026. Final grade 9.61/10.
- Supervisor: Jerffeson Teixeira de Souza. Co-supervisor: Allysson Allex Araújo.
- CAPES-funded visiting PhD researcher at Brunel University London, UK (Oct 2024 - Apr 2025), hosted by Dr. Giuseppe Destefanis.

MSc in Computer Science, State University of Ceará (UECE), Brazil — Mar 2019 - Sep 2021

- Dissertation: *A multi-method sociotechnical study for the development and evaluation of a blockchain-based Electronic Patient Record* (in Portuguese). Final grade 9.34/10.
- Supervisor: Jerffeson Teixeira de Souza. Co-supervisor: Allysson Allex Araújo.

BSc in Computer Science, State University of Ceará (UECE), Brazil — 2013 - Dec 2018

- Final project on a multi-objective evolutionary algorithm based on preference-driven polyhedral cones (N2PC).

RESEARCH EXPERIENCE

Doctoral Researcher (CAPES scholarship), UECE, Fortaleza — Mar 2022 - Feb 2026

- Developed BcDEx, a multidimensional framework for blockchain developer experience, combining a multivocal literature review, repository mining, developer interviews and a survey (15 interviews, 60 survey responses).
- Built a prototype LLM pipeline that classifies GitHub issues into a four-level developer-experience taxonomy (three chained prompts with JSON output; API-hosted and local models via Ollama); analysed a 1,835-issue sample with Kruskal-Wallis, Dunn/Bonferroni and Mann-Whitney U tests and measured inter-model agreement with Gwet's AC1.
- Co-built Web3BlockSet (MSR 2026), a public dataset of 391,596 GitHub issues and pull requests from 381 official and 4,500 community repositories; categories proposed with an LLM and validated by four researchers (92.4% agreement).
- Led an empirical study of Ethereum ERC standard adoption in open source projects (SBES 2026).

Visiting PhD Researcher (CAPES), Brunel University London, UK — Oct 2024 - Apr 2025

- Hosted by Giuseppe Destefanis. Cleaned and topic-modelled 12,000+ GitHub issues and pull requests from nine Blockchain-as-a-Service platforms (BERTopic; c_v coherence of about 0.49 to 0.50); GPT-4 proposed topic labels reviewed by three researchers; emotion analysis with a pre-trained RoBERTa GoEmotions model; code and data shared on Zenodo.
- Co-authored *Blockchain Developer Experience: A Multivocal Literature Review* (IEEE/ACM CHASE 2025) with UK-based collaborators.

Researcher and Team Lead, IDESCO (Blockchain One and Urbe.Digital projects), Fortaleza — May 2020 - Feb 2024

- Led the research and architecture of DocStone, a blockchain-based document registration service (Third Best Paper Award, SBCARS 2022; extended in JSERD 2023).
- Contributed to research on integrating federal property systems with blockchain for process transparency (WCGE 2023).

Master's Researcher (CAPES scholarship), UECE — 2019 - 2021

- Designed and evaluated a blockchain-based Electronic Patient Record using sociotechnical, user-centred methods (published in JSERD 2025); built a microservices and blockchain prototype for notary offices (ICSA-C 2020).

Researcher, CryptumLab, UECE — 2018 - 2020

Undergraduate Research Fellow (PIBIC/CNPq), GOES Lab, UECE — 2017 - 2019

- Applied multi-objective evolutionary algorithms (NSGA-II) to software release planning, with controlled experiments and statistical analysis.

TEACHING AND SUPERVISION

- **Supervised teaching practice (postgraduate programme, UECE):** taught the statistics module of the Experimental Software Engineering course, covering descriptive statistics and hypothesis testing for empirical studies.

- **Short course:** "Descriptive Statistics and Hypothesis Testing" (2021).
- **Supervision:** co-supervised undergraduate final-year projects (TCC) and undergraduate research (scientific initiation) on blockchain topics at UECE, and mentored students in research and development projects.
- **Examination:** member of undergraduate final-year project examination panels at UECE and the Federal University of Ceará (UFC).
- **PET tutorial programme (UECE, 2014 - 2017):** teaching support in Analytic Geometry and Introduction to Computer Science, and short courses for undergraduate students.

FUNDING, AWARDS AND HONOURS

- CAPES doctoral scholarship, 2022 - 2026.
- CAPES scholarship for the visiting PhD period at Brunel University London, 2024 - 2025.
- CAPES master's scholarship, 2019 - 2020.
- FUNCAP Technological Innovation Grant (BIT), UX/UI and quality assurance for the Ceará Valley Digital Platform, 2026 - present.
- CNPq undergraduate research scholarship (PIBIC), 2017 - 2019; PET tutorial programme scholarship, 2014 - 2017.
- Third Best Paper Award, Brazilian Symposium on Software Components, Architectures, and Reuse (SBCARS 2022), for DocStone.
- Second place, Research Design track, XVII Brazilian Symposium on Collaborative Systems (SBSC 2021).
- Led team to 1st place (Challenge 3, Contract Management) and 3rd place (Challenge 1, Characterisation and Incorporation) at the Web3 Hackathon on Tokenisation of Federal Assets, Brazilian Federal Property Secretariat (SPU), 2022.
- Third place, II Hackathon do Conhecimento, 2018.

SERVICE AND OUTREACH

- **Reviewer:** WETSEB 2026 (ICSE workshop); CBSoft 2026 Artifacts Festival; Brazilian Symposium on Computers in Education (SBIE) 2024 and 2025; ISE 2025; International Congress on Blockchain and Applications (BLOCKCHAIN) 2023 and 2024.
- **Organisation:** coordinator of the UECE Computing Week (SECOMP) 2016 and organising committee member 2014, 2015 and 2017; coordinator of the Ceará Informatics Olympiad (OCI) 2015.
- **Outreach:** speaker at the round table "Sandwich PhD in England: research and lessons learned" (2025). **Research software and data:** Web3BlockSet dataset and dashboard (2026); BcDEx web diagnostic tool (2026).

PUBLICATIONS

Publications in Portuguese keep their original titles.

Journal articles

[1] **Soares, Pamella**; Araújo, Allysson Allex; Saraiva, Raphael; Souza, Jerffeson. Bridging Blockchain Technology to Electronic Patient Record Engineering: A Sociotechnical and User-centered Analysis. Journal of Software Engineering Research and Development, v. 13, 2025.

[2] **Soares, Pamella**; Saraiva, R. L.; Fernandes, I.; Araujo, Allysson Allex; Souza, J. T.; Loiola, Ricardo. Extending the Docstone to Enable a Blockchain-based Service for Customizable Assets and Blockchain Types. Journal of Software Engineering Research and Development, v. 11, 2023.

[3] Yeltsin, Italo; Araújo, Allysson Allex; Dantas, Altino; **Soares, Pamella**; Saraiva, Raphael; Souza, Jerffeson. On the -fairness- of search-based software engineering: Investigating the capability of a scalarizing-based function to control the software metrics' influence on optimization process. Journal of Software Engineering Research and Development, v. 13, 2025.

[4] Câmara, Samuel; Allex Araújo, Allysson; Buarque, Brenno; **Soares, Pamella**; Queiroz, Letícia; Souza, Jerffeson. Will the Innovation Drivers be Shaped by the Metaverses? A Theoretical Essay Based on a Systematic Mapping Study. International Journal of Innovation and Technology Management, v. 20, 2023.

[5] Araujo, Allysson Allex; **Soares, Pamella**; Saraiva, Raphael; Camara, Samuel; Souza, Jerffeson. Dominating the wild: How have companies been leveraging Cryptoeconomics to shape markets through Business Model Innovation? International Journal on Management of Innovation & Technology, v. 1, 2022.

[6] Rocha, Filipe Herculano; Araújo, Allysson Allex; **Soares, Pamella**; Saraiva, Raphael Lima; De Souza, Jerffeson Teixeira. Uma Avaliação de Desempenho de Soluções Off-chain baseadas em Sistemas de Armazenamento Distribuído. iSys - Revista Brasileira de Sistemas de Informação, v. 14, 2021.

[7] Saraiva, R. L.; Allysson Allex; **Sousa, P. S.**; Souza, J. T. MIRIAM: proposta de solução baseada em blockchain para gerenciamento de registros profissionais de médicos. Revista de Saúde Digital e Tecnologias Educacionais (RESDITE), v. 6, 2021.

[8] Saraiva, Raphael; Câmara, Samuel; Souza, Jerffeson; Araújo, Allysson Allex; **Soares, Pamella**. Cryptoeconomics as a Market Shaping Phenomenon upon Business Model Innovation: A Systematic Literature Review. International Journal of Business Innovation and Research, v. 1, 2021.

Conference and workshop papers

[9] **Soares, Pamella**; Destefanis, G.; dos Santos, Allan; Allysson Allex; Saraiva, Raphael; Souza, Jerffeson. Web3BlockSet: A Dataset for Empirical Research in Blockchain-Oriented Software Engineering. In: International Conference on Mining Software Repositories (MSR 2026), Data and Tool Showcase Track, 2026, Rio de Janeiro.

[10] **Soares, Pamella**; Silva Filho, A.; Rodrigues, R.; Araújo, Allysson; Souza, J. T. From Specification to Practice: An Empirical Study of Ethereum ERC Standard Adoption in Open Source Projects. In: Simpósio Brasileiro de Engenharia de Software (SBES 2026), 2026, São Paulo.

[11] **Soares, Pamella**; Araújo, Allysson Allex; Destefanis, Giuseppe; Neykova, Rumyana; Saraiva, Raphael; Souza, Jerffeson. Blockchain Developer Experience: A Multivocal Literature Review. In: 2025 IEEE/ACM 18th International Conference on Cooperative and Human Aspects of Software Engineering (CHASE), 2025, Ottawa.

[12] **Soares, Pamella**; Araújo, Allysson Allex; Saraiva, Raphael; Souza, Jerffeson. Towards Blockchain Developer Experience (BcDEx): Exploring Dimensions of Developer Experience in Blockchain-oriented Software Engineering. In: Simpósio Brasileiro de Engenharia de Software, 2024, Brasil.

[13] **Soares, Pamella**; Saraiva, Raphael; Fernandes, Iago; Neto, Antonio; Souza, Jerffeson. A Blockchain-based Customizable Document Registration Service for Third Parties. In: 2022 IEEE International Conference on Blockchain and Cryptocurrency (ICBC), 2022, Shanghai.

[14] **Soares, Pamella**; Saraiva, Raphael; Fernandes, Iago; Souza, Jerffeson; Loiola, Ricardo. DocStone: A Blockchain-based Architecture for a Customizable Document Registration Service. In: SBCARS 2022: 16th Brazilian Symposium on Software Components, Architectures, and Reuse, 2022, Uberlandia Brazil.

[15] **Sousa, P. S.**; Allysson Allex; Saraiva, R. L.; Santos, Rayane Celestino dos; Souza, J. T. Prontuário Eletrônico do Paciente baseado em blockchain: Um Desenho de Pesquisa Sociotécnico. In: XVII Simpósio Brasileiro de Sistemas Colaborativos, 2021.

[16] **Soares, Pamella**; Allysson Allex; Saraiva, R. L.; Souza, Jerffeson. Escalabilidade no contexto de Prontuário Eletrônico do Paciente baseado em Blockchain: Um Estudo Experimental sobre Armazenamento Off-chain. In: III Workshop de Modelagem e Simulação de Sistemas Intensivos em Software (MSSiS), 2021, Joinville, SC.

[17] **Soares, Pamella**; Allysson Allex; Saraiva, R. L.; Souza, Jerffeson; Sousa, G. J. B.; Pereira, M. L. D. Prontuário Eletrônico do Paciente baseado em Blockchain: Uma Análise das Potencialidades e Desafios à Luz dos Requisitos da SBIS e LGPD. In: XII Congresso Brasileiro de Software: Teoria e Prática (CBSoft) - Trilha da Indústria, 2021, Joinville, SC.

[18] **Sousa, Pamella Soares de**; Nogueira, Nataniel Parente; Santos, Rayane Celestino dos; Maia, Paulo Henrique M.; Souza, Jerffeson Teixeira de. Building a prototype based on Microservices and Blockchain technologies for notary's office: An academic experience report. In: 2020 IEEE International Conference on Software Architecture Companion (ICSA-C), 2020, Salvador.

[19] **Sousa, P. S.**; Saraiva, R. L.; Allysson Allex; Campos, G.; Souza, J. T. N2PC: Um Algoritmo Evolucionário Multiobjetivo baseado em Cones Poliédricos construídos através de Preferências. In: IX Workshop de Engenharia de Software Baseada em Busca (WESB 2018), 2018, São Carlos.

[20] Fernandes, Iago; **Soares, Pamella**; Saraiva, Raphael; Vasconcelos, Ana Klyssia M.; Paixao, Matheus; Araújo, Allysson Allex; Souza, Jerffeson. Adoption of Blockchain for Decentralized Execution Agreements: Empirical Evidence of Usability and Acceptance. In: Workshop sobre Aspectos Sociais, Humanos e Econômicos de Software, 2026, Brasil.

[21] Saraiva, Raphael; Araújo, Allysson Allex; Santos, Ronnie de Souza; **Soares, Pamella**; Souza, Jerffeson. Towards an Intelligent Platform for Supporting Quality Assessment in Blockchain Systems. In: 8th International Workshop on Emerging Trends in Software Engineering for Blockchain (WETSEB 2026), co-located with ICSE 2026, 2026, Rio de Janeiro.

[22] Saraiva, Raphael; Araújo, Allysson Allex; **Soares, Pamella**; Pontes, João Carlos; Souza, Jerffeson. Metrics for Quality Assessment in Blockchain-based Systems: A Systematic Mapping Study. In: Simpósio Brasileiro de Sistemas de Informação, 2025, Brasil.

[23] Costa, Saori; Paixao, Matheus; Steinmacher, Igor; **Soares, Pamella**; Araújo, Allysson Allex; Souza, Jerffeson. Sociotechnical Dynamics in Open Source Smart Contract Repositories: An Exploratory Data Analysis of Curated High Market Value Projects. In: PROMISE '24: 20th International Conference on Predictive Models and Data Analytics in Software Engineering, 2024, Porto de Galinhas Brazil.

[24] Saraiva, Raphael; **Soares, Pamella**; Araújo, Allysson Allex; Souza, Jerffeson. Análise descritiva dos microdados do Censo da Educação Superior do INEP para cursos de Tecnologia da Informação e Comunicação no Brasil - Um panorama 2015-2021. In: Workshop sobre Educação em Computação, 2023, Brasil.

[25] Fernandes, Iago; **Soares, Pamella**; Saraiva, Raphael; Vicente, Bruno; Sertã, Alexandre; Loiola, Ricardo; Araújo, Allysson; Souza, Jerffeson. Uma proposta de integração de sistemas da Secretaria do Patrimônio da União ao blockchain para incentivo à transparência de processos: resultados de uma pesquisa em andamento. In: Workshop de Computação Aplicada em Governo Eletrônico, 2023, Brasil.

[26] Damasceno, A.; **Soares, Pamella**; Santos, I.; Souza, J. T.; Oliveira, F. Assistive Technology for Distance Education in Metaverse-Based Environment: A Rapid Review. In: XXXIV Simpósio Brasileiro de Informática na Educação, 2023, Passo Fundo/RS.

[27] Gomes, Francisco; **Soares, Pamella**; Araújo, Allysson Allex. Examining the Performance Implications of Design Patterns in Front-end Web Development: A Preliminary Comparative Study with React and Vue.js. In: WebMedia '23: Brazilian Symposium on Multimedia and the Web, 2023, Ribeirão Preto Brazil.

[28] Jefferson Macedo, Antonio; Allex Araújo, Allysson; Saraiva, Raphael; **Soares, Pamella**; Tomaz, Antonio Emerson B.; Taveira, Itamara. Towards a blockchain-based API to ensure data interoperability and transparency in the registration and inspection processes of Brazilian water dams. In: SBSI '23: XIX Brazilian Symposium on Information Systems, 2023, Maceió Brazil.

[29] Santos, Rayane Celestino dos; **Soares, Pamella**; Rodrigues, E.; Silveira, A.; Maia, P. H. M. How Blockchain and Microservices are Being Used Together: a Systematic Mapping Study. In: 5th International Workshop on Emerging Trends in Software Engineering for Blockchain (WETSEB), 2022, Pittsburgh.

[30] Rodrigues, Alan; Araújo, Allysson Allex; Paixão, Matheus; **Soares, Pamella**. Aspectos Colaborativos no Desenvolvimento de Contratos Inteligentes da Plataforma Ethereum: Um Estudo Exploratório-descritivo Preliminar. In: Workshop sobre Aspectos Sociais, Humanos e Econômicos de Software, 2022, Brasil.

[31] Alves, Gabriela; **Soares, Pamella**; Souza, Bianca; Veras, Necio; Araújo, Allysson Allex. What is students' perception of Flipped Classroom usage during Emergency Remote Teaching of Software Engineering subjects? An Experience Report in Brazil. In: SBES 2022: XXXVI Brazilian Symposium on Software Engineering, 2022, Virtual Event Brazil.

[32] Castro, R.; Allysson Allex; **Soares, Pamella**; Paixao, M.; Maia, P. H. M.; Souza, J. T. Proposta de Mapa Interativo e Colaborativo para Ecossistemas de Inovação: Um Estudo de Caso Baseado no Estado do Ceará. In: XVII Simpósio Brasileiro de Sistemas Colaborativos, 2021.

[33] Saraiva, Raphael; Araujo, Allysson Allex; **Soares, Pamella**; Souza, Jerffeson. MIRIAM: A Blockchain-based Web Application for Managing Professional Registrations of Medical Doctors in Brazil. In: 2021 IEEE International Conference on Blockchain and Cryptocurrency (ICBC), 2021, Sydney.

[34] Rodrigues, A.; Araújo, Allysson Allex; Paixao, M.; **Soares, Pamella**. Caracterizando a evolução de software de contratos inteligentes: Um estudo exploratório-descritivo utilizando GitHub e Etherscan. In: IX Workshop de Visualização, Evolução e Manutenção de Software, 2021, Joinville, SC.

[35] Teotonio, W. C.; Allysson Allex; **Soares, Pamella**; Souza, J. T.; Paixao, M. Murphy: Um Estudo de Caso utilizando Unity para Desenvolvimento de Jogos Casuais Mobile. In: XIX Simpósio Brasileiro de Games e Entretenimento Digital, 2020.

[36] Cortés, Mariela; Saraiva, Raphael; Souza, Marcia; Mello, Patricia; **Soares, Pamella**. Adoption of Software Testing in Internet of Things. In: IV Brazilian Symposium on Systematic and Automated Software Testing (SAST 2019), 2019, Salvador.

[37] Souza, J. T.; Allysson Allex; Saraiva, R. L.; **Sousa, P. S.**; Maia, C. A Preliminary Systematic Mapping Study of Human Competitiveness of SBSE. In: 10th International Symposium on Search Based Software Engineering (SSBSE 2018), 2018, Montpellier.

[38] Souza, J. T.; Allysson Allex; Yeltsin, I.; Saraiva, R. L.; **Sousa, P. S.**. On the Placebo Effect in Interactive SBSE: A Preliminary Study. In: 10th International Symposium on Search Based Software Engineering (SSBSE 2018), 2018, Montpellier.

Interdisciplinary collaboration (Wi-Fi sensing)

[39] Costa Nascimento dos Santos, Allan; **Soares, Pamella**; Galdino, Iandra; Ramos, Taiane C.; Albuquerque, Célio; Guerra, Raphael; Fernandes, Natalia C.; Muchaluat-Saade, Débora C.; Ghinea, Gheorghita. A transformer-based framework for device-free human activity recognition using Wi-Fi CSI. Annals of Telecommunications, v. 80, 2026.

[40] dos Santos, Allan Costa Nascimento; **Soares, Pamella**; Galdino, Iandra; Soto, Julio C. H.; De Sousa, Cledson; Ramos, Taiane C.; De Albuquerque, Celio V. N.; Guerra, Raphael; Fernandes, Natalia C.; Muchaluat-Saade, Débora C.; Ghinea, Gheorghita. A Transformer-Based Methodology for Person-Independent Human Activity Recognition Using Wi-Fi CSI. In: 2025 13th Wireless Days Conference (WD), 2025, Niterói.

INDUSTRY EXPERIENCE

Blockchain Developer | Smart Contract Engineer, Bitshopp, São Paulo (remote) Feb 2024 - Present

- Enterprise tokenisation platform: Solidity smart contracts (ERC-20, ERC-721, ERC-1155), RESTful APIs integrated with Hyperledger Besu, static analysis with Slither and Mythril. Took part in the ANBIMA pilot on tokenisation of debentures and investment funds (with Laqus).

Blockchain Developer and Team Lead, IDESCO (Blockchain One and Urbe.Digital projects) May 2020 - Feb 2024

- Led the team that delivered Sis.integra, the first municipal blockchain solution for license and permit validation; Node.js/Express.js APIs integrating blockchain services with government systems.

SKILLS

- **ML / NLP / LLM:** BERTopic, Sentence-Transformers, Hugging Face Transformers (inference), PyTorch, NLTK; LLMs via API (GPT-4, Claude, Mistral) and locally (Ollama).

- **Data and statistics:** Python, pandas, NumPy, SciPy, statsmodels, scikit-posthocs, matplotlib; non-parametric tests, inter-rater agreement, mixed methods; LaTeX.
- **Blockchain and Software engineering:** Solidity, TypeScript/JavaScript, Node.js, SQL (MySQL, PostgreSQL), Git, GitHub API.
- **Languages:** Portuguese (native), English (B2, upper intermediate).

REFEREES

- **Jerffeson Teixeira de Souza**, Professor, State University of Ceará (UECE), Brazil. PhD supervisor.
- **Giuseppe Destefanis, Professor, University College London (UCL), United Kingdom. g.destefanis@ucl.ac.uk**

Full contact details for both referees available on request.